Dynamic-Equation and Analysis Contract for an IEEE 14-Bus Mixed-Resource System

One EMF6 Synchronous Generator and Four 10-State GFL–RMS10 Converters

Power Flow, Equilibrium, Full-State SSSA, and Event-Free Time-Domain Simulation

In-house Power-Flow Project

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Abstract

This report documents the exact positive-sequence RMS equations used by the project for the IEEE 14-bus mixed-resource case. The fixed operating mode contains SG1 at bus 1 and IBR2, IBR3, IBR6, and IBR8 in 10-state PLL-resolved grid-following (GFL) mode. The report defines every state, reference frame, sign convention, per-unit conversion, network residual, equilibrium condition, SSSA reduction, and implicit trapezoidal time step. The reported operating-point tables, SSSA, and the 15-s event-free TS all use the same accepted SG1 plus four-GFL equilibrium. Numerical plots are rendered natively by LaTeX from CSV files exported from the project-owned MATLAB solvers, so the plot font, size, and spacing are identical to the report. Numerical instability or fail-closed outcomes are reported rather than tuned away.

Contents

1 Scope, evidence discipline, and reproducibility

The production calculation uses only project-owned base-MATLAB code. The allowed numerical primitives include `\` and `eig`; no external power-flow, optimization, Simulink, or loaded-solution result feeds the calculation. The evidence files in `\datadir` are generated by `scripts/reporting/generate_ieee14_ibr_pf_sssa_ts_report_data.m`. That script calls the same production dispatch used by `solve_case`, writes the full-precision results to CSV and MAT files, and records branch, commit, profile, convergence, and physical classifications in `manifest.txt`.

The products in this report are deliberately separated:

1. mode-aware all-GFL PF warm-start with effective PQ converter buses;
2. mixed-device equilibrium and full-state SSSA at the fixed four-GFL mode map;
3. a 15-s event-free normal-operation TS from that equilibrium.

2 Case data and index maps

2.1 System and resource map

The network is the 100-MVA, 60-Hz IEEE 14-bus data set. The external bus ID is not assumed equal to an arbitrary internal index, even though this case happens to use consecutive IDs. Algebraic voltage order is

$$\boldsymbol{y} = [V_{x,1}, V_{y,1}, V_{x,2}, V_{y,2}, \ldots, V_{x,14}, V_{y,14}]^{\mathsf{T}}, \qquad V_k = V_{x,k} + jV_{y,k}.$$

The fixed normal resource map is

Device index	Bus ID	Device	Normal mode	Physical states	Active in normal SSSA
1	1	SG1	synchronous EMF6	6	5
2	2	IBR2	GFL-RMS10	10 of dual 23	10
3	3	IBR3	GFL-RMS10	10 of dual 23	10
4	6	IBR6	GFL-RMS10	10 of dual 23	10
5	8	IBR8	GFL-RMS10	10 of dual 23	10
			Active dynamic order	$5 + 4(10) = 45$	

The runtime inventory remains fixed at 98 states: SG1 contributes 6 and each dual-capable IBR reserves 23 coordinates. For this report, only the ten GFL-RMS10 coordinates of each IBR enter the 45-state reduced dynamic matrix.

2.2 Bus and line input data

Table 1: Case bus data. These are inputs, not solved PF results.

Bus	Type	P_L MW	Q_L MVAr	V_0 pu	θ_0 deg	Base kV	V_{max}	V_{min}
1	REF	0	0	1.06	0	0	1.06	0.94
2	PV	21.7	12.7	1.045	-4.98	0	1.06	0.94
3	PV	94.2	19	1.01	-12.72	0	1.06	0.94
4	PQ	47.8	-3.9	1.019	-10.33	0	1.06	0.94
5	PQ	7.6	1.6	1.02	-8.78	0	1.06	0.94
6	PV	11.2	7.5	1.07	-14.22	0	1.06	0.94
7	PQ	0	0	1.062	-13.37	0	1.06	0.94
8	PV	0	0	1.09	-13.36	0	1.06	0.94
9	PQ	29.5	16.6	1.056	-14.94	0	1.06	0.94
10	PQ	9	5.8	1.051	-15.1	0	1.06	0.94
11	PQ	3.5	1.8	1.057	-14.79	0	1.06	0.94
12	PQ	6.1	1.6	1.055	-15.07	0	1.06	0.94
13	PQ	13.5	5.8	1.05	-15.16	0	1.06	0.94
14	PQ	14.9	5	1.036	-16.04	0	1.06	0.94

Table 2: Case branch series and charging data in project index order.

Index	From	To	r pu	x pu	b_c pu
1	1	2	0.01938	0.05917	0.0528
2	1	5	0.05403	0.22304	0.0492
3	2	3	0.04699	0.19797	0.0438
4	2	4	0.05811	0.17632	0.034
5	2	5	0.05695	0.17388	0.0346
6	3	4	0.06701	0.17103	0.0128
7	4	5	0.01335	0.04211	0
8	4	7	0	0.20912	0
9	4	9	0	0.55618	0
10	5	6	0	0.25202	0
11	6	11	0.09498	0.1989	0
12	6	12	0.12291	0.25581	0
13	6	13	0.06615	0.13027	0
14	7	8	0	0.17615	0
15	7	9	0	0.11001	0
16	9	10	0.03181	0.0845	0
17	9	14	0.12711	0.27038	0
18	10	11	0.08205	0.19207	0
19	12	13	0.22092	0.19988	0
20	13	14	0.17093	0.34802	0

Table 3: Case branch ratings, transformer ratios, phase shifts, and status in the same index order.

Index	From	To	Rate A	Tap	Shift deg	Status
1	1	2	0	0	0	1
2	1	5	0	0	0	1
3	2	3	0	0	0	1
4	2	4	0	0	0	1
5	2	5	0	0	0	1
6	3	4	0	0	0	1
7	4	5	0	0	0	1
8	4	7	0	0.978	0	1
9	4	9	0	0.969	0	1
10	5	6	0	0.932	0	1
11	6	11	0	0	0	1
12	6	12	0	0	0	1
13	6	13	0	0	0	1
14	7	8	0	0	0	1
15	7	9	0	0	0	1
16	9	10	0	0	0	1
17	9	14	0	0	0	1
18	10	11	0	0	0	1
19	12	13	0	0	0	1
20	13	14	0	0	0	1

The source bus-data base-kV column is zero (unspecified). The project case contract supplies a 69-kV reporting base, so solved values use $V_{\text{kV}} = 69|V|_{\text{pu}}$; this reporting conversion does not enter the per-unit PF, equilibrium, SSSA, or TS equations.

For a branch $i \to j$, $z_{ij} = r_{ij} + jx_{ij}$, $y_{ij} = 1/z_{ij}$, and the charging susceptance is divided equally between the terminal shunts. Tap magnitude τ and phase shift ϕ enter the off-nominal transformer terms. These contributions assemble the complex network matrix Y.

3 Common frames, signs, and per-unit conversion

All devices use generator sign: positive complex power is injected into the network,

$$S = P + jQ = VI^*, \qquad I_{bus} = \sum_{d \in bus} I_d.$$

For an IBR whose local angle is δ_c, the network-to-device transform is

$$v_d = V_x \cos\delta_c + V_y \sin\delta_c, \qquad v_q = -V_x \sin\delta_c + V_y \cos\delta_c, \tag{1}$$

$$i_x = i_d \cos\delta_c - i_q \sin\delta_c, \qquad i_y = i_d \sin\delta_c + i_q \cos\delta_c. \tag{2}$$

Thus $v_q = 0$ when a PLL is locked to the positive-sequence bus voltage. With this convention,

$$P_{inv} = v_d i_d + v_q i_q, \qquad Q_{inv} = v_q i_d - v_d i_q.$$

If S_{base} is the system base and M_{base} is the converter base,

$$\kappa = \frac{S_{base}}{M_{base}}, \qquad P_{inv} = \kappa P_{sys}, \quad Q_{inv} = \kappa Q_{sys}, \quad I_{sys} = \frac{I_{inv}}{\kappa}.$$

IBR2 uses $M_{base} = 140$ MVA, hence $\kappa = 100/140$; IBR3, IBR6, and IBR8 use 100 MVA and $\kappa = 1$.

4 Composite network DAE

Every device supplies a differential right-hand side and a positive complex current injection. The composite equations are

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma), \tag{3}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma) = \mathcal{R}(Y(\sigma)V - I_{bus}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma)), \tag{4}$$

where $\mathcal{R}(z) = [\Re z_1, \Im z_1, \ldots]^\mathsf{T}$ and σ contains topology, online status, and device mode. At a declared reference bus, the appropriate KCL rows are replaced by explicit voltage constraints; the replacement is metadata-driven, not inferred inside a device.

5 Synchronous generator: EMF6 equations

5.1 State order and meaning

$$\boldsymbol{x}_{SG} = [\delta, \omega, E'_q, E'_d, E''_q, E''_d]^\mathsf{T}, \qquad \boldsymbol{u}_{SG} = [T_m, E_{fd}]^\mathsf{T}.$$

Local	State	Unit	Physical meaning
1	δ	rad	electrical rotor angle in the network reference frame
2	ω	pu	rotor-speed deviation; actual speed is $\omega_b(1+\omega)$
3	E'_q	pu	q-axis transient internal EMF
4	E'_d	pu	d-axis transient EMF; frozen at zero here because $T'_{q0} = 0$
5	E''_q	pu	q-axis subtransient internal EMF
6	E''_d	pu	d-axis subtransient internal EMF

The physical inventory is sixth order, but the normal SSSA contribution is five because the sourced Kodsi round-rotor data impose the singular limit $T'_{q0} = 0 \Rightarrow E'_d = 0$.

5.2 Stator algebraic equations and ODEs

The project Kundur transform gives (V_d, V_q) from (V_x, V_y, δ). Defining $r_d = V_d - E''_d$, $r_q = V_q - E''_q$, and $\Delta = R_a^2 + X''_d X''_q$, the stator current is

$$I_d = \frac{-R_a r_d - X''_q r_q}{\Delta}, \qquad I_q = \frac{X''_d r_d - R_a r_q}{\Delta}, \tag{5}$$

$$T_e = V_d I_d + V_q I_q + R_a(I_d^2 + I_q^2). \tag{6}$$

On the system base, the connected-machine dynamics are

$$\dot{\delta} = \omega_b \omega, \tag{7}$$

$$2H_{sys}\dot{\omega} = T_m - T_e - D_{sys}\omega, \tag{8}$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \tag{9}$$

$$T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d, \tag{10}$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \tag{11}$$

$$T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q. \tag{12}$$

When the SG breaker is open, $I_d = I_q = T_e = 0$. The angle and speed coast, and the four flux equations use the same formulas with their current terms removed. The retained T_m policy is therefore visible in the post-trip trajectory; it is not silently re-solved at every time step.

6 GFM–VSG REGFM_B1: 13-state equations

These equations document the reserved grid-forming branch of each dual-mode IBR. That branch is inactive throughout the operating point studied in this report: it contributes no row to the 45-state SSSA matrix and is not integrated by the event-free TS. It is retained here to make the complete device model and inactive-state inventory auditable.

6.1 State order and device frame

$$\boldsymbol{x}_{GFM} = [\omega_m, \delta_{IT}, x_w, x_E, \delta_{PLL}, \xi_{PLL}, P_f, I_{d,f}, Q_f, V_f, I_{q,f}, \delta_{IT}^{max}, \delta_{IT}^{min}]^\mathsf{T}. \tag{13}$$

Local	State	Meaning
1	ω_m	VSG speed deviation
2	δ_{IT}	PLL-relative VSG internal angle
3	x_w	damping washout state
4	x_E	voltage-controller integral state
5–6	δ_{PLL}, ξ_{PLL}	PLL angle and PI integral
7–11	$P_f, I_{d,f}, Q_f, V_f, I_{q,f}$	measured active power, d current, reactive power, voltage and q current filters
12–13	$\delta_{IT}^{max}, \delta_{IT}^{min}$	dynamic upper/lower current-angle bounds

The device voltage angle is

$$\delta_{VSM} = \delta_{PLL} + \text{clamp}\big(\delta_{IT}, \delta_{IT}^{min}, \delta_{IT}^{max}\big), \quad \text{clamp}\big(z, l, h\big) = \min\big(\max(z, l), h\big).$$

The clamp is an algebraic projection: an interior value is unchanged, a value below l is replaced by l, and a value above h is replaced by h. The derivative uses an outward-hold/inward-release rule rather than differentiating through an impossible outward motion.

6.2 PLL, VSG, voltage and filter dynamics

With $V_q = -V_x \sin \delta_{PLL} + V_y \cos \delta_{PLL}$,

$$\dot{\xi}_{PLL} = V_q, \qquad \dot{\delta}_{PLL} = \omega_b (k_{p,PLL} V_q + k_{i,PLL} \xi_{PLL}), \tag{14}$$

$$2H\dot{\omega}_m = P_{ref,inv} - P_f - (1/m_p + D_1)\omega_m - D_2(\omega_m - x_w), \tag{15}$$

$$\dot{x}_w = \omega_D(\omega_m - x_w), \tag{16}$$

$$\dot{\delta}_{IT} = \text{conditional\_hold}\Big(\omega_b\omega_m - \dot{\delta}_{PLL}; \delta_{IT}^{min}, \delta_{IT}^{max}\Big), \tag{17}$$

$$\dot{x}_E = \text{conditional\_hold}_E(V_{ref} - V_f), \tag{18}$$

$$E_{VSM}^{raw} = V_{ref} - m_q Q_f + k_{pv}(V_{ref} - V_f) + k_{iv} x_E. \tag{19}$$

The voltage-integrator hold is active only when the voltage command is at a limit and the error would drive it farther outward. The measurement states are

$$T_{pf}\dot{P}_f = \kappa P - P_f, \qquad T_{If}\dot{I}_{d,f} = \kappa I_d - I_{d,f}, \tag{20}$$

$$T_{Qf}\dot{Q}_f = \kappa Q - Q_f, \qquad T_{Vf}\dot{V}_f = |V| - V_f, \tag{21}$$

$$T_{If}\dot{I}_{q,f} = \kappa I_q - I_{q,f}. \tag{22}$$

The moving angle bounds satisfy

$$\dot{\delta}_{IT}^{max} = \text{conditional\_hold}\big(k_I(I_{d,max}^{SS} - I_{d,f}); 0, \delta_{max}\big), \tag{23}$$

$$\dot{\delta}_{IT}^{min} = \text{conditional\_hold}\big(k_I(-K_e I_{d,max}^{SS} - I_{d,f}); -\delta_{max}, 0\big), \tag{24}$$

with the lower state frozen when its source flag is disabled.

The unsaturated network current is the voltage-behind-impedance relation

$$I_{unc} = \frac{E_{VSM} e^{j\delta_{VSM}} - V}{\kappa(R_e + jX_L)}.$$

Its magnitude is circularly limited at I_{maxF}/κ, after the sourced P/Q and voltage-limit logic. This current, not a separate SSSA surrogate, is injected into the common KCL equations.

7 GFL–RMS10: 10-state PLL-resolved equations

7.1 State order and meaning

$$\boldsymbol{x}_{GFL} = [\delta_{PLL}, \xi_{PLL}, P_f, Q_f, \xi_P, \xi_Q, \xi_{id}, \xi_{iq}, i_d, i_q]^{\mathsf{T}}.$$

Local	State	Unit	Meaning
1	δ_{PLL}	rad	SRF-PLL angle used by every dq transform
2	ξ_{PLL}	pu s	PLL PI integral of v_q
3–4	P_f, Q_f	pu inv.	filtered measured injected powers
5–6	ξ_P, ξ_Q	pu s	outer active/reactive-power PI integrals
7–8	ξ_{id}, ξ_{iq}	pu s	inner d/q current-controller PI integrals
9–10	i_d, i_q	pu inv.	physical L-filter current states in the PLL frame

The last two rows are why this model is a genuine dynamic GFL rather than a PLL label attached to algebraic current commands.

7.2 PLL, power measurement, and exact P/Q feedforward

Let $D_V = v_d^2 + v_q^2$. The declared positive-sequence domain requires $D_V \geq V_{div,min}^2$. Within that domain,

$$\dot{\xi}_{PLL} = v_q, \qquad \Delta\omega_{PLL} = k_{p,PLL} v_q + k_{i,PLL}\xi_{PLL}, \qquad \dot{\delta}_{PLL} = \omega_b \Delta\omega_{PLL}, \tag{25}$$

$$T_P \dot{P}_f = P_{inv} - P_f, \qquad T_Q \dot{Q}_f = Q_{inv} - Q_f. \tag{26}$$

The exact inverse of the dq power equations supplies the steady feedforward:

$$i_{d,ff} = \frac{v_d P_{ref} + v_q Q_{ref}}{D_V}, \qquad i_{q,ff} = \frac{v_q P_{ref} - v_d Q_{ref}}{D_V}, \tag{27}$$

$$i_{d,r}^{raw} = k_{pP}(P_{ref} - P_f) + k_{iP}\xi_P + i_{d,ff}, \tag{28}$$

$$i_{q,r}^{raw} = -k_{pQ}(Q_{ref} - Q_f) - k_{iQ}\xi_Q + i_{q,ff}. \tag{29}$$

The minus sign in the q-axis controller follows the generator convention $Q = v_q i_d - v_d i_q$: at a locked PLL, positive reactive injection requires negative i_q.

7.3 Current priority, balanced LVRT, and outer anti-windup

In normal voltage, circular P-priority limits $\left(i_{d,r}^{raw}, i_{q,r}^{raw}\right)$ to I_{max}: i_d is clipped first and i_q receives the remaining circle. During the declared balanced positive-sequence voltage dip,

$$i_{q,FRT} = \mathrm{clamp}\big(K_{qv}\,\mathrm{max}(V_{dip} - |V| - d_b, 0), -I_{qh1}, I_{qh1}\big), \tag{30}$$

$$i_{d,LVPL}^{max} = \begin{cases} 0, & |V| \leq Z_{erox}, \\ L_{vpl1}\dfrac{|V| - Z_{erox}}{B_{rkpt} - Z_{erox}}, & Z_{erox} < |V| < B_{rkpt}, \\ L_{vpl1}, & |V| \geq B_{rkpt}, \end{cases} \tag{31}$$

$$|i_d| \leq \mathrm{min}\left(i_{d,LVPL}^{max}, \sqrt{I_{max}^2 - i_q^2}\right). \tag{32}$$

This is reactive-current priority during the voltage dip; it does not cover negative sequence or zero-voltage ride through.

After limiting, $P_{lim} = v_d i_{d,r} + v_q i_{q,r}$ and $Q_{lim} = v_q i_{d,r} - v_d i_{q,r}$. The outer integrators are

$$\dot{\xi}_P = AW_P(P_{ref} - P_f), \qquad \dot{\xi}_Q = AW_Q(Q_{ref} - Q_f). \tag{33}$$

Each AW returns zero only when its limiter is active and its error would push the command farther outward; inward recovery remains active. The comparison tolerance is numerical metadata, not a tunable physical gain.

7.4 Inner current PI, voltage command, and L-filter ODE

With $e_d = i_{d,r} - i_d$, $e_q = i_{q,r} - i_q$, and $\omega_{PLL,pu} = 1 + \Delta\omega_{PLL}$,

$$v_{td}^{raw} = k_{pi}e_d + k_{ii}\xi_{id} + R_t i_d - \omega_{PLL,pu} L i_q + v_d, \tag{34}$$

$$v_{tq}^{raw} = k_{pi}e_q + k_{ii}\xi_{iq} + R_t i_q + \omega_{PLL,pu} L i_d + v_q. \tag{35}$$

The vector $\left[v_{td}^{raw}, v_{tq}^{raw}\right]$ is projected onto the modulation circle of radius $V_{t,max} = m_{max}V_{dc0}$. The current PI integrators use the same outward-hold/inward-release anti-windup rule:

$$\dot{\xi}_{id} = AW_{id}(e_d), \qquad \dot{\xi}_{iq} = AW_{iq}(e_q).$$

Because L is stored as per-unit reactance, the per-unit inductance in the derivative denominator is L/ω_b:

$$\dot{i}_d = \frac{\omega_{PLL,pu} L i_q - R_t i_d + v_{td} - v_d}{L/\omega_b}, \tag{36}$$

$$\dot{i}_q = \frac{-\omega_{PLL,pu} L i_d - R_t i_q + v_{tq} - v_q}{L/\omega_b}. \tag{37}$$

Finally,

$$I_{net} = \frac{(i_d + j i_q) e^{j\delta_{PLL}}}{\kappa}, \qquad S = V I_{net}^*.$$

8 How PF, equilibrium, SSSA, and TS use the equations

8.1 Power flow

PF does not integrate converter or SG states. For each non-reference bus it solves the specified active/reactive mismatch selected by the REF/PV/PQ semantics. In compact form, Newton iterations solve

$$J_{PF}(z^{(m)})\Delta z^{(m)} = -F_{PF}(z^{(m)}), \qquad z^{(m+1)} = z^{(m)} + \Delta z^{(m)},$$

until $\|F_{PF}\|_\infty$ is below tolerance. The source-case REF/PV/PQ PF is used only to freeze the constant-admittance load model. The reported operating point instead uses the mode-aware warm-start, changes every online GFL terminal to its effective PQ semantics, and then accepts only the subsequent coupled equilibrium satisfying the full device ODEs and every KCL row. Thus all reported bus, load, branch, and device tables share the same voltage vector and SG1 plus four-GFL mode map.

8.2 Mixed-device equilibrium

Equilibrium uses the full device equations and solves

$$\boldsymbol{f}_A(\boldsymbol{x}_0, \boldsymbol{y}_0, \boldsymbol{u}_0) = 0, \qquad \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{y}_0, \boldsymbol{u}_0) = 0,$$

for active states and algebraic variables. Frozen or inactive branch states remain in the inventory but not in the Newton unknown vector. GFL lock gives $v_q = 0$, $P_f = P_{ref}$, $Q_f = Q_{ref}$, and stationary current states. SG equilibrium solves constant T_m, E_{fd}, which are then held in TS.

8.3 Full-state SSSA

Linearizing the same DAE at a fixed active regime gives

$$\begin{bmatrix} \Delta\dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}, \qquad A_{full} = f_x - f_y(g_y \backslash g_x).$$

The published matrix is $A = A_{full}(\mathcal{A}, \mathcal{A})$, where $\mathcal{A}$ is the metadata-owned active-state map. No eigenvalue is manually deleted. For $Av_i = \lambda_i v_i$ and $A^H u_i = \lambda_i^* u_i$, the vectors are paired and normalized so that $u_i^H v_i = 1$. Signed participation is $p_{ki} = u_{ki}^* v_{ki}$; the displayed nonnegative ranking is $\rho_{ki} = |p_{ki}| / \sum_j |p_{ji}|$. Mode number, raw eigen index, state index, and device index are separate quantities.

8.4 Time-domain simulation

Between event transactions, the implicit trapezoidal step solves

$$0 = x_{n+1} - x_n - \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right], \tag{38}$$

$$0 = g(x_{n+1}, y_{n+1}; \sigma_{n+1}). \tag{39}$$

The public sampling interval is $h = 0.01$ s. This report uses an empty event schedule, so the fixed all-GFL mode map remains unchanged for all 1500 steps.

9 All-GFL equilibrium and SSSA results

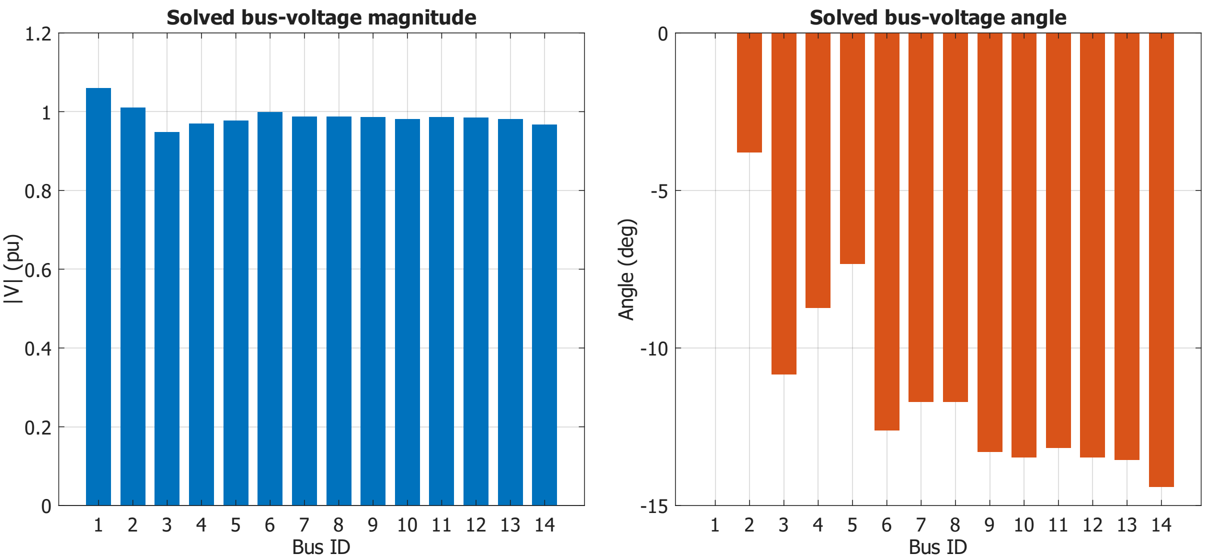

Figure 1: Accepted SG1 plus four-GFL equilibrium bus-voltage results.

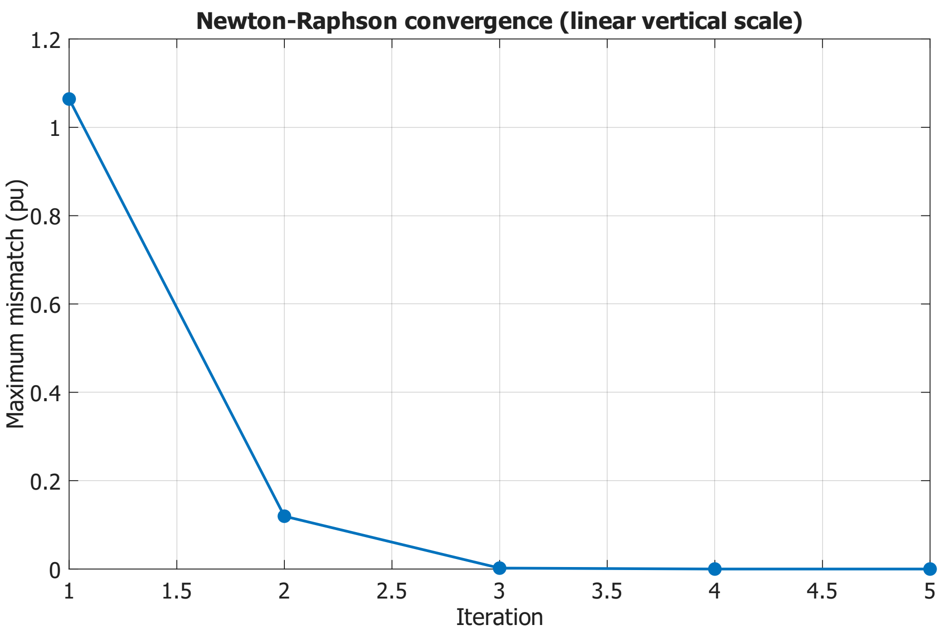

Figure 2: Mode-aware all-GFL PF warm-start convergence on a linear vertical scale.

Table 4: Solved bus voltages at the accepted SG1 plus four-GFL equilibrium.

Bus	Type	$\lvert V\rvert$ pu	$\lvert V\rvert$ kV	δ deg
1	REF	1.06	73.14	0
2	PQ	1.0095	69.66	−3.7885
3	PQ	0.9482	65.42	−10.835
4	PQ	0.9695	66.9	−8.719
5	PQ	0.9775	67.45	−7.3352
6	PQ	0.9986	68.91	−12.6144
7	PQ	0.9869	68.1	−11.7171
8	PQ	0.9869	68.1	−11.7171
9	PQ	0.9857	68.01	−13.3028
10	PQ	0.9811	67.69	−13.4668
11	PQ	0.9865	68.07	−13.172
12	PQ	0.9848	67.95	−13.4669
13	PQ	0.9804	67.65	−13.5457
14	PQ	0.9666	66.69	−14.4089

Table 5: All-GFL equilibrium bus generation, voltage-dependent admittance load, and net injection. All quantities are on the 100-MVA system base and positive net injection denotes power delivered to the network.

Bus	P_{gen} pu	Q_{gen} pu	P_{load} pu	Q_{load} pu	P_{net} pu	Q_{net} pu
1	2.0171	0.7125	0	0	2.0171	0.7125
2	0.4	0	0.2025	0.1185	0.1975	−0.1185
3	0	0	0.8302	0.1675	−0.8302	−0.1675
4	0	0	0.4338	−0.0354	−0.4338	0.0354
5	0	0	0.0699	0.0147	−0.0699	−0.0147
6	0	0	0.0976	0.0653	−0.0976	−0.0653
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0.2571	0.1447	−0.2571	−0.1447
10	0	0	0.0784	0.0505	−0.0784	−0.0505
11	0	0	0.0305	0.0157	−0.0305	−0.0157
12	0	0	0.0531	0.0139	−0.0531	−0.0139
13	0	0	0.1176	0.0505	−0.1176	−0.0505
14	0	0	0.1298	0.0436	−0.1298	−0.0436

Table 6: All-GFL equilibrium branch sending-end flows and losses. Power quantities are on the 100-MVA system base; the sign follows the listed from–to orientation.

Line	From	To	P_{ij} pu	Q_{ij} pu	P_{loss} pu	Q_{loss} pu
1	1	2	1.3584	0.4695	0.036125	0.053729
2	1	5	0.6587	0.243	0.024384	0.049512
3	2	3	0.64	0.175	0.02068	0.045117
4	2	4	0.5045	0.066	0.014911	0.01194
5	2	5	0.3753	0.0563	0.008175	−0.009201
6	3	4	−0.2109	−0.0376	0.003392	−0.003113
7	4	5	−0.5449	−0.0046	0.004218	0.013304
8	4	7	0.2447	0.0272	0	0.012899
9	4	9	0.1417	0.0323	0	0.011738
10	5	6	0.3824	0.2264	0	0.045239
11	6	11	0.063	0.0311	0.00047	0.000985
12	6	12	0.0677	0.0218	0.000624	0.001298
13	6	13	0.1541	0.0629	0.001837	0.003618
14	7	8	0	0	0	0
15	7	9	0.2447	0.0143	0	0.006786
16	9	10	0.0466	0.0367	0.000115	0.000306
17	9	14	0.0828	0.0314	0.001025	0.002181
18	10	11	−0.032	−0.0142	0.000104	0.000244
19	12	13	0.0139	0.0066	0.000054	0.000049
20	13	14	0.0485	0.0153	0.00046	0.000937

Table 7: Solved device injections at the same all-GFL mixed-device equilibrium.

Index	Device	Bus	Mode	P MW	Q MVAr	$\|V\|$ pu	$\|V\|$ kV
1	SG1	1	SG	201.708	71.248	1.06	73.14
2	IBR2	2	GFL	40	0	1.0095	69.656
3	IBR3	3	GFL	0	0	0.9482	65.425
4	IBR6	6	GFL	0	0	0.9986	68.907
5	IBR8	8	GFL	0	0	0.9869	68.097

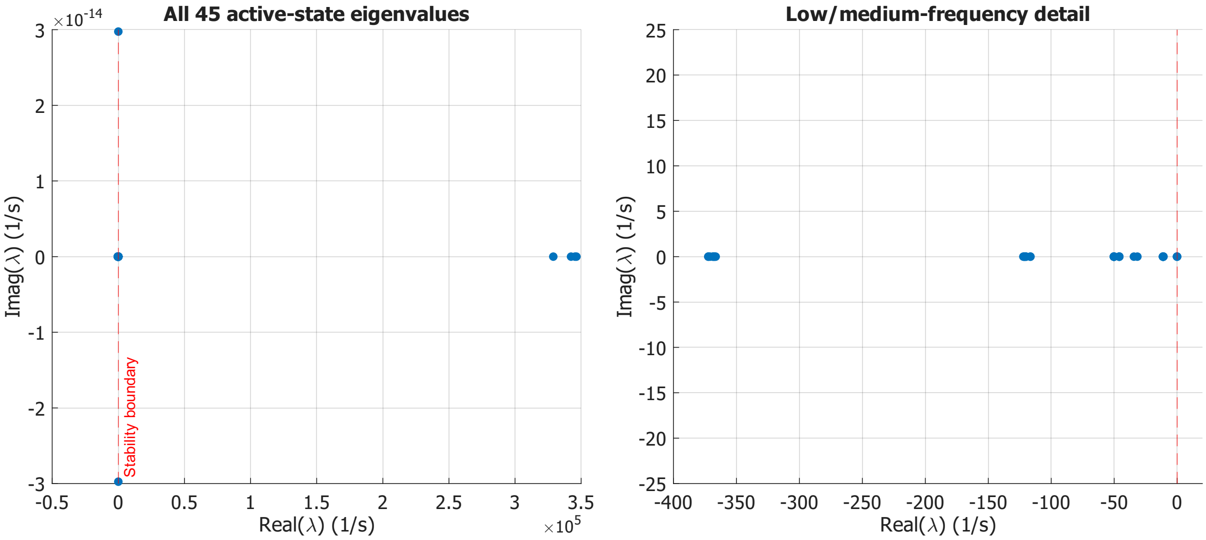

Figure 3: All 45 SG1 plus four-GFL eigenvalues (left) and the low/medium-frequency detail (right). Each point is one eigenvalue of the same 45-state reduced matrix. The dashed line is $\Re\lambda = 0$; roots to its right are unstable.

9.1 SG1 plus four GFL–RMS10 normal-operation SSSA

The explicit all-GFL mode map keeps SG1 online and assigns IBR2, IBR3, IBR6, and IBR8 to GFL–RMS10. Its independently solved equilibrium has 45 active states. The complete spectrum below is published without filtering: the four large positive real PLL roots and the small positive SG angle root make the physical result **UNSTABLE**. Successful equilibrium and SSSA execution therefore must not be interpreted as a stable classification.

Table 8: Complete 45-root SSSA result for SG1 plus four GFL–RMS10 resources.

No.	Raw	$\Re\lambda$ (s^{-1})	$\Im\lambda$ (s^{-1})	f (Hz)	ζ	Device	Local state	Part. (%)
1	4	$3.464 \cdot 10^{5}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	-1	IBR6	14	99.99
2	3	$3.453 \cdot 10^{5}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	-1	IBR2	14	99.98
3	2	$3.423 \cdot 10^{5}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	-1	IBR8	14	99.99
4	1	$3.289 \cdot 10^{5}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	-1	IBR3	14	99.99
5	11	$4.834 \cdot 10^{-4}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	-1	SG1	1	99.38
6	12	$-1.485 \cdot 10^{-1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	3	90.67
7	13	$-1.924 \cdot 10^{-1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	2	95.44
8	29	$-1.093 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
9	28	$-1.093 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
10	38	$-1.116 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
11	43	$-1.116 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
12	36	$-1.123 \cdot 10^{1}$	$2.975 \cdot 10^{-14}$	$4.735 \cdot 10^{-15}$	1	SG1	1	NaN
13	37	$-1.123 \cdot 10^{1}$	$-2.975 \cdot 10^{-14}$	$4.735 \cdot 10^{-15}$	1	SG1	1	NaN
14	23	$-1.129 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	IBR2	19	76.59
15	22	$-1.129 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
16	14	$-3.161 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	6	86.36
17	15	$-3.437 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	5	83.42
18	18	$-4.599 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
19	19	$-4.599 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
20	21	$-4.599 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
21	20	$-4.599 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	IBR2	15	99.99
22	30	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
23	31	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
24	45	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
25	44	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
26	35	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
27	34	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
28	27	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
29	26	$-5.023 \cdot 10^{1}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
30	16	$-1.166 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
31	17	$-1.166 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
32	42	$-1.201 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
33	41	$-1.201 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
34	33	$-1.212 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
35	32	$-1.212 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
36	8	$-1.216 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	IBR2	16	53.15
37	7	$-1.222 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	IBR2	17	52.86
38	6	$-3.665 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	IBR2	23	70.7
39	24	$-3.674 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
40	25	$-3.674 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
41	39	$-3.685 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
42	40	$-3.685 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
43	5	$-3.709 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	IBR2	22	70.99
44	10	$-3.723 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN
45	9	$-3.723 \cdot 10^{2}$	$0.000 \cdot 10^{0}$	$0.000 \cdot 10^{0}$	1	SG1	1	NaN

10 Event-free time-domain result

The SG1 plus four-GFL normal-operation TS starts from the same converged equilibrium used by SSSA and completes all 1500 fixed 0.01-s steps to 15 s. The empty event schedule keeps SG1 online and all four IBRs in GFL-RMS10 mode. No fault, breaker transaction, or automatic mode switch is included.

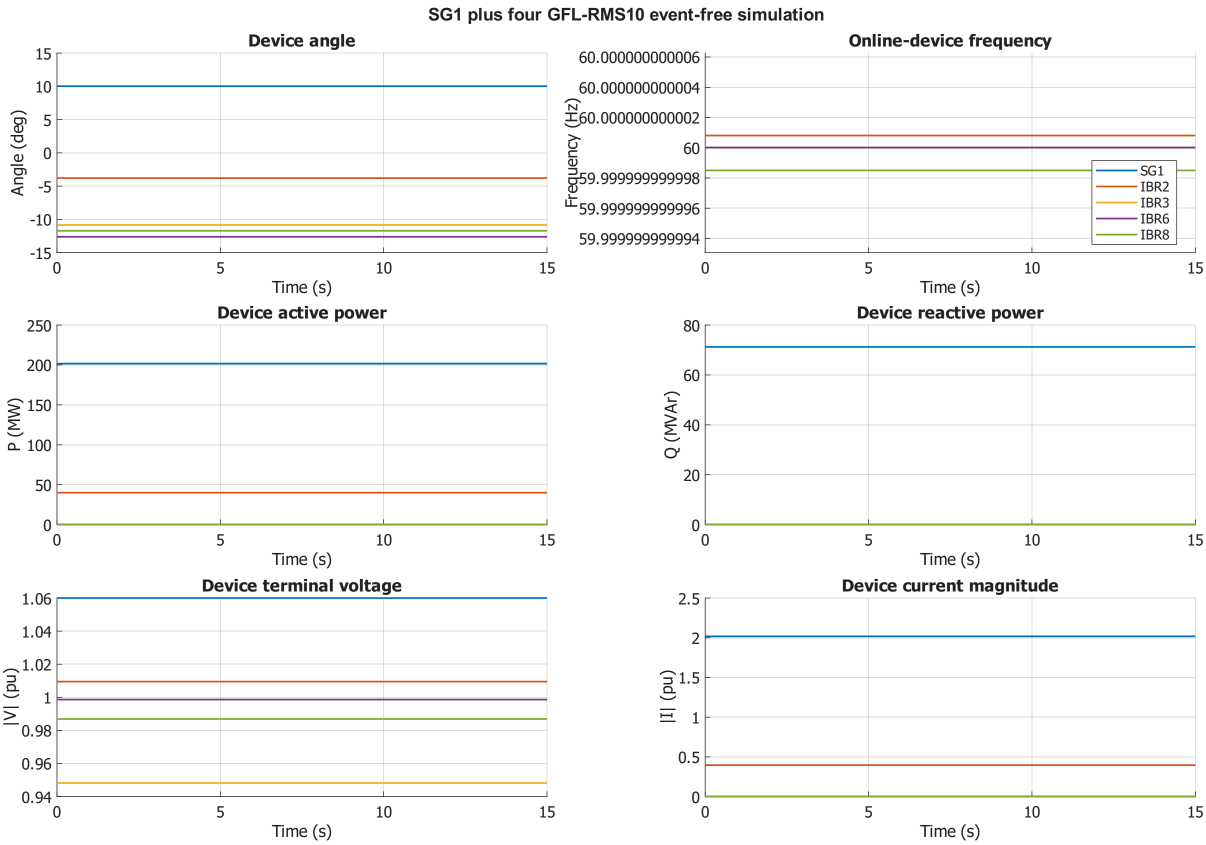

Figure 4: MATLAB event-free TS for SG1 plus four GFL-RMS10 resources over 15 s.

11 Comparison with the conventional SG analysis path

The conventional multi-SG path and the mixed IBR path share the network and analysis mathematics, but not the device states. A classical SG contributes $[\delta, \omega]$, while the present SG contributes the EMF6 inventory and the converter models contribute control and physical current states. PF is common because all resources appear as scheduled injections and voltage constraints. Equilibrium, SSSA, and TS are generic consumers of the device ABI: they do not contain a device-specific eigenvalue matrix. In the present fixed mode map, the four IBRs contribute only their GFL-RMS10 coordinates and current laws to the Schur-complement and implicit-trapezoidal algorithms.

12 Limitations and interpretation

- The GFL fault contract is balanced positive-sequence RMS LVRT. It is not an EMT, negative-sequence, or zero-voltage model.
- A completed numerical SSSA execution is distinct from a stable physical classification. Positive-real-part eigenvalues remain in the table and plot.

- The documented grid-forming branch remains inactive in every numerical result in this report; its equations are model documentation, not evidence that any IBR was assigned that mode.

13 Conclusion

The report exposes a traceable chain from case data through device equations, state order, initialization, DAE assembly, PF, SSSA, and TS. The same nonlinear SG and four GFL-RMS10 equations initialize the equilibrium, are differentiated for SSSA, and are integrated for TS. Native LaTeX plots retain the exact MATLAB series while enforcing common typography and symmetric layout. The inactive grid-forming equations remain documented without being mixed into the reported numerical operating point.

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